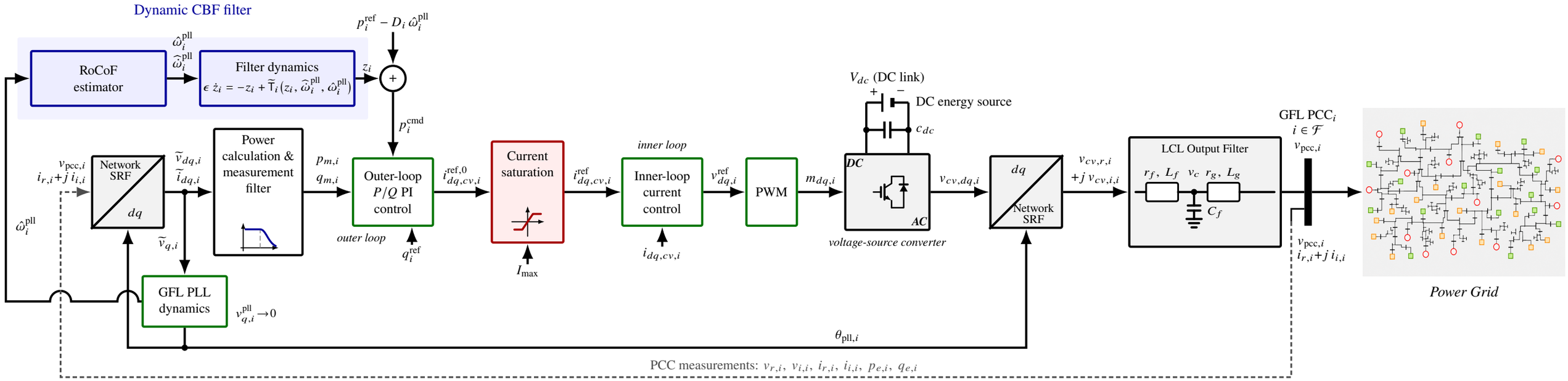

Dynamic CBF filter
RoCoF estimator
Filter dynamics
$\epsilon\,\dot{z}_i = -z_i + \bar{\mathsf{T}}_i(z_i, \hat{\omega}_i^{\mathrm{pll}}, \hat{\omega}_i^{\mathrm{pll}})$
$\hat{\omega}_i^{\mathrm{pll}}$
$\widehat{\omega}_i^{\mathrm{pll}}$
z_i
$p_i^{\mathrm{ref}} - D_i\,\hat{\omega}_i^{\mathrm{pll}}$
p_i^{cmd}
$v_{\mathrm{pcc},i}$
$i_{r,i}+j\,i_{i,i}$
Network SRF
dq
$\widetilde{v}_{dq,i}$
$\widetilde{i}_{dq,i}$
$\widetilde{v}_{q,i}$
$\hat{\omega}_i^{\mathrm{pll}}$
Power calculation & measurement filter
$p_{m,i}$
$q_{m,i}$
Outer-loop P/Q PI control
outer loop
q_i^{ref}
$i_{dq,cv,i}^{\mathrm{ref},0}$
Current saturation
$I_{\max}$
$i_{dq,cv,i}^{\mathrm{ref}}$
inner loop
Inner-loop current control
$i_{dq,cv,i}$
$v_{dq,i}^{\mathrm{ref}}$
PWM
$m_{dq,i}$
V_{dc} (DC link)
+
−
DC energy source
c_{dc}
DC
AC
voltage-source converter
$v_{cv,dq,i}$
dq
Network SRF
$v_{cv,r,i}$
$+j\,v_{cv,i,i}$
LCL Output Filter
$r_f,\ L_f$
v_c
$r_g,\ L_g$
C_f
GFL PCC$_i$
$i \in \mathcal{F}$
$v_{\mathrm{pcc},i}$
$v_{\mathrm{pcc},i}$
$i_{r,i}+j\,i_{i,i}$
Power Grid
GFL PLL dynamics
$v_{q,i}^{\mathrm{pll}} \to 0$
$\theta_{\mathrm{pll},i}$
PCC measurements: $v_{r,i},\ v_{i,i},\ i_{r,i},\ i_{i,i},\ p_{e,i},\ q_{e,i}$